LAB 5

By

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Date (Month, day, year assignment is due or turned in if late)

“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to--except as documented in a bibliography or acknowledged below.

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Artificial Intelligence (AI) Acknowledgement: I used ChatGPT to assist in drafting some repetitive graphical analysis after showing it how I wanted them to be done based on the first few. I also used it to draft a conclusion and an introduction.

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1. Contents

List of Figures 1
List of Tables 2
Executive Summary 3
Introduction 4
Methods 4
Results 5
Benchmark for Total Parts 5
Graphical Analysis 5
Numerical Summary 8
Predictions of Future Performance 10
Benchmark for Good Parts 10
Graphical Analysis 10
Numerical Summary 13
Predictions of Future Performance 14
Check Sheet and Pareto Chart Analysis 15
Benchmark of Off-spec Parts 17
Graphical Analysis 17
Numerical Summary 21
Predictions of Future Performance 23
Benchmark of Bad Parts 23
Graphical Analysis 23
Numerical Summary 27
Predictions of Future Performance 29
Conclusions 29
Recommendations 29

List of Figures

Figure 1. Run Chart of Total Parts 5
Figure 2. Histogram of Total Parts 6
Figure 3. Boxplot of Total Parts 7
Figure 4. Probability Plot of Total Parts 8
Figure 5. I-MR Chart of Total Parts 9

Figure 6. Run Chart of Good Parts 10
Figure 7. Histogram of Total Parts 11
Figure 8. Box Plot of Good Parts 12
Figure 9. Probability Plot of Good Parts 13
Figure 10. I-MR Chart of Good Parts 14
Figure 11. Pareto Chart of Nonconforming Parts 17
Figure 12. Run Chart of Off-spec Parts 18
Figure 13. Histogram of Off-spec Parts 19
Figure 14. Box Plot of Off-spec Parts 20
Figure 15. Probability Plot of Off-spec Parts 21
Figure 16. I-MR Chart of Off-spec Parts 22
Figure 17. Run Chart of Bad Parts 24
Figure 18. Histogram of Bad Parts 25
Figure 19. Box Plot of Bad Parts 26
Figure 20. Probability Plot of Bad Parts 27
Figure 21. I-MR Chart of Bad Parts 28

List of Tables

Table 1. Descriptive Statistics of Total Parts 8
Table 2. Descriptive Statistics of Good Parts 13
Table 3. Check Sheet of Quality Nonconformities 15
Table 4. Descriptive Statistics of Off-spec Parts 22
Table 5. Descriptive Statistics of Bad Parts 27

2. Executive Summary

This report evaluates 20 days of production and quality data from the Mouse Factory to assess current process performance and identify the primary sources of nonconforming output. Benchmark analyses of the total, good, off-spec, and bad daily parts show that the process is stable and that the average daily production statistics correspond to nearly 272,000 parts per million nonconforming. Out of the 69 observed types of nonconformity, 6 types accounted for 76.3%, and 3 types accounted for 54.9%. Reducing variation and mistake proofing should be directed towards these areas while under tight statistical process control for the most efficient process improvement.

3. Introduction

The Mouse Factory produces USB optical mice and maintains daily production and quality records, tracking the number of conforming and nonconforming parts. As the newly assigned quality manager, the objective of this analysis is to evaluate 20 days of historical production data to assess current process performance capability and identify the major sources of quality loss. Parts that fall outside specification limits for a critical quality characteristic are classified as off-spec, while parts that fail inspection for another reason are classified as bad parts. Parts meeting all requirements are counted as good parts. These categories, along with total daily output, provide the basis for developing production and quality benchmarks used to identify improvement opportunities and support targeted recommendations.

4. Methods

To assess process capability, a benchmark analysis will be conducted for total parts to evaluate the stability of daily production volume, and for good parts to understand the factory's effective productive output. To assess nonconformities in the process, additional benchmark analyses will be conducted for off-spec and bad parts. Each benchmark will consist of a graphical analysis, a numerical summary of performance measures, and predictions of future performance.

Graphical analyses will consist of a run chart to illustrate process behavior over time and detect non-random patterns, a boxplot to summarize the central tendency, spread, and potential outliers, a histogram to display the overall shape of the distribution, and a probability plot to assess whether the data follow a normal distribution.

The numerical summary will provide several descriptive statistics, including the mean, standard deviation, and a 95% confidence interval calculated using ***Equation 1*** for each group of data.

$$\bar{X} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}} \tag{1}$$

$\bar{X}$ is the sample mean across 20 days, t is the critical value from the t-distribution at a confidence level of 95% (α=0.05), s is the sample standard deviation, n is the sample size. To determine whether the variation and mean are stable, an I–MR chart will be included. This control chart was selected because the data consist of daily individual observations rather than subgroups.

Predictions of future performance will be made by calculating a prediction interval for each set of data using ***Equation 2***.

$$\bar{X} - t_{\frac{\alpha}{2},n-1} s \sqrt{1 + \frac{1}{n}} < X_{n+1} < \bar{X} + t_{\frac{\alpha}{2},n-1} s \sqrt{1 + \frac{1}{n}} \tag{2}$$

X_{n+1} is the next individual observation. This interval quantifies the range of values in which a future daily observation is expected to fall with 95% confidence and provides a forward-looking assessment of expected process behavior.

To establish recommendations for process improvement, ***Equation 3*** will be used to find the parts per million nonconforming.

$$\frac{\bar{X}_{off-spec} + \bar{X}_{bad}}{\bar{X}_{total}} \times 1{,}000{,}000 \tag{3}$$

5. Results

5.1. Benchmark for Total Parts

5.1.1. Graphical Analysis

Figure 1 shows a run chart of the daily total parts to evaluate whether the process exhibits non-random patterns over time.

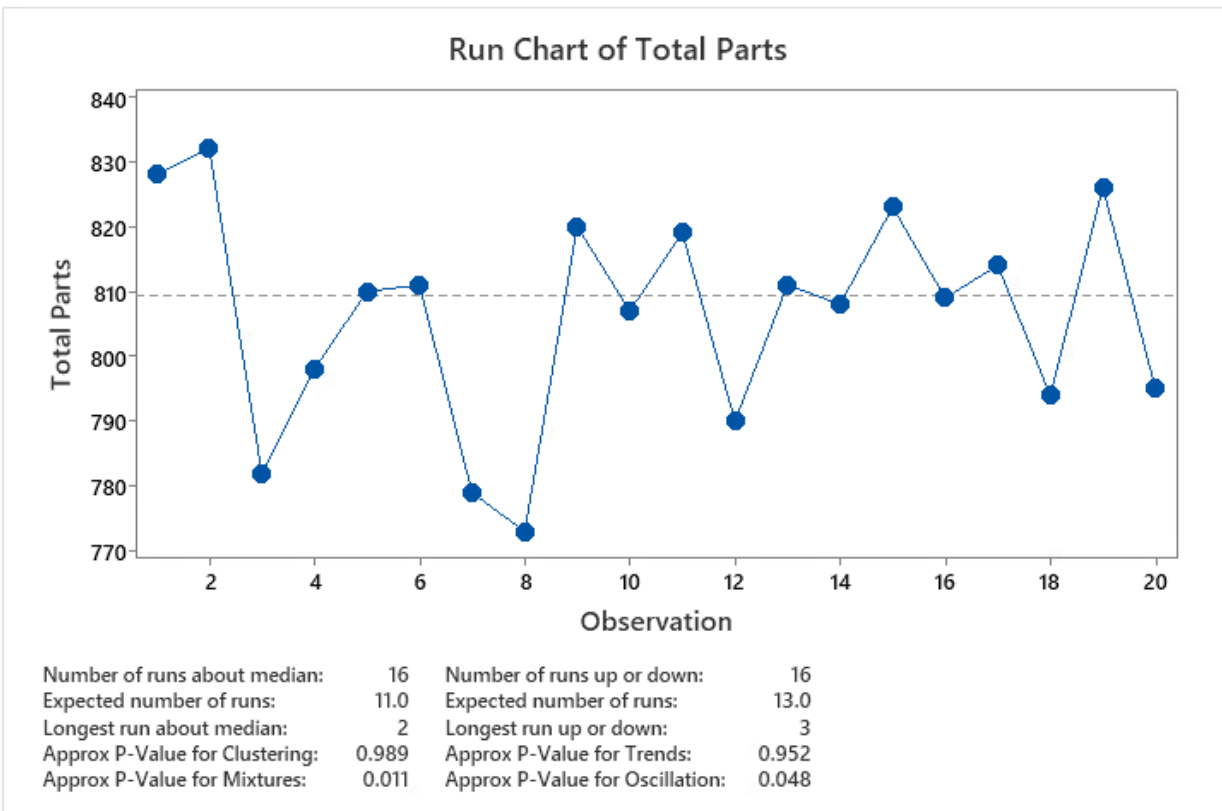

FIGURE 1. RUN CHART OF TOTAL PARTS

The observations fluctuate around the median without extended runs, shifts, or trends. The number of runs is close to what would be expected under randomness, and the p-values for clustering and trend do not indicate non-random behavior. Although the mixture p-value is low, the chart does not display a meaningful structural pattern, and the overall production levels remain within a stable range. These results suggest that total daily output remained steady throughout the 20-day period.

Figure 2 shows a histogram of the daily total parts to illustrate how the observations are distributed across their range of values.

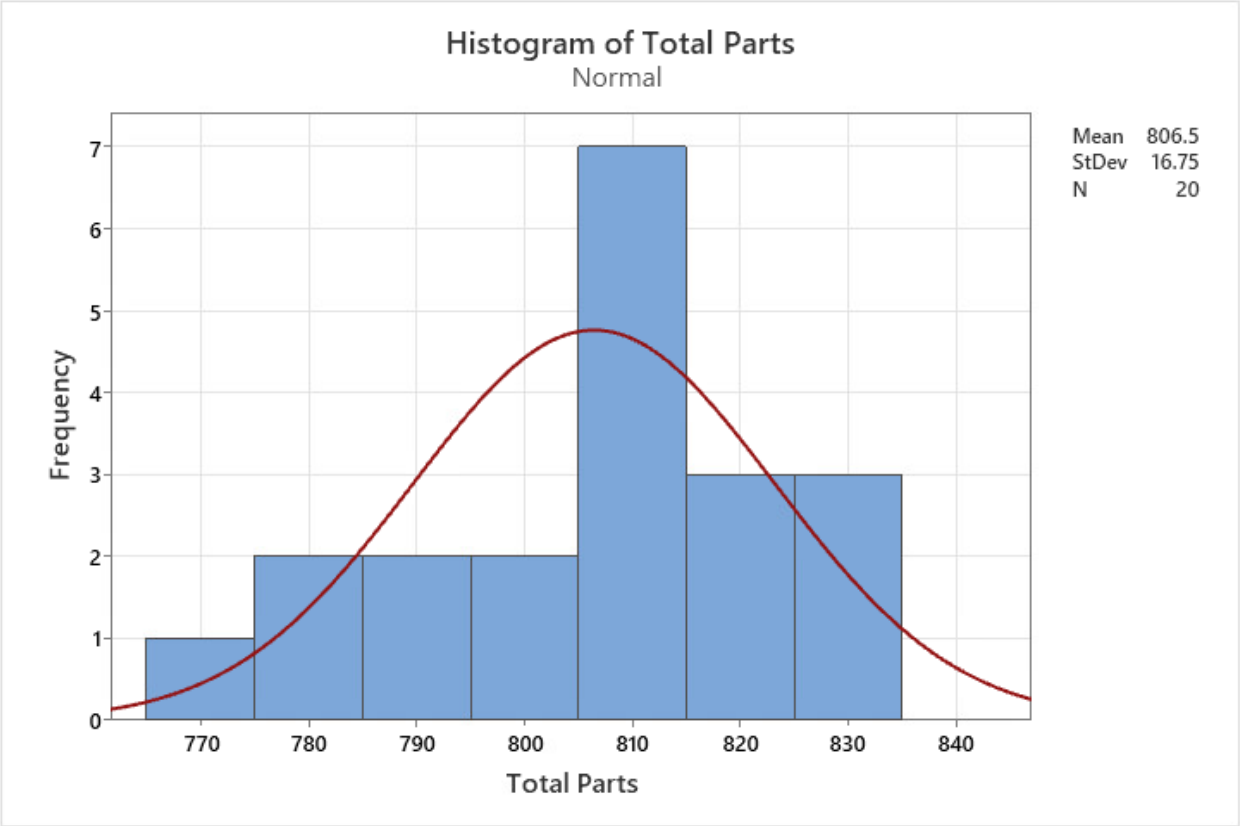

FIGURE 2. HISTOGRAM OF TOTAL PARTS

Daily output appears roughly symmetric around the mean, with no extreme values or noticeable skewness. The fitted normal curve aligns closely with the distribution, suggesting that total production varies in a predictable pattern around its central value. The overall spread indicates moderate day-to-day fluctuation in production volume without unusually high or low output days.

Figure 3 shows a boxplot of the daily total parts to summarize the central tendency and variability of the data.

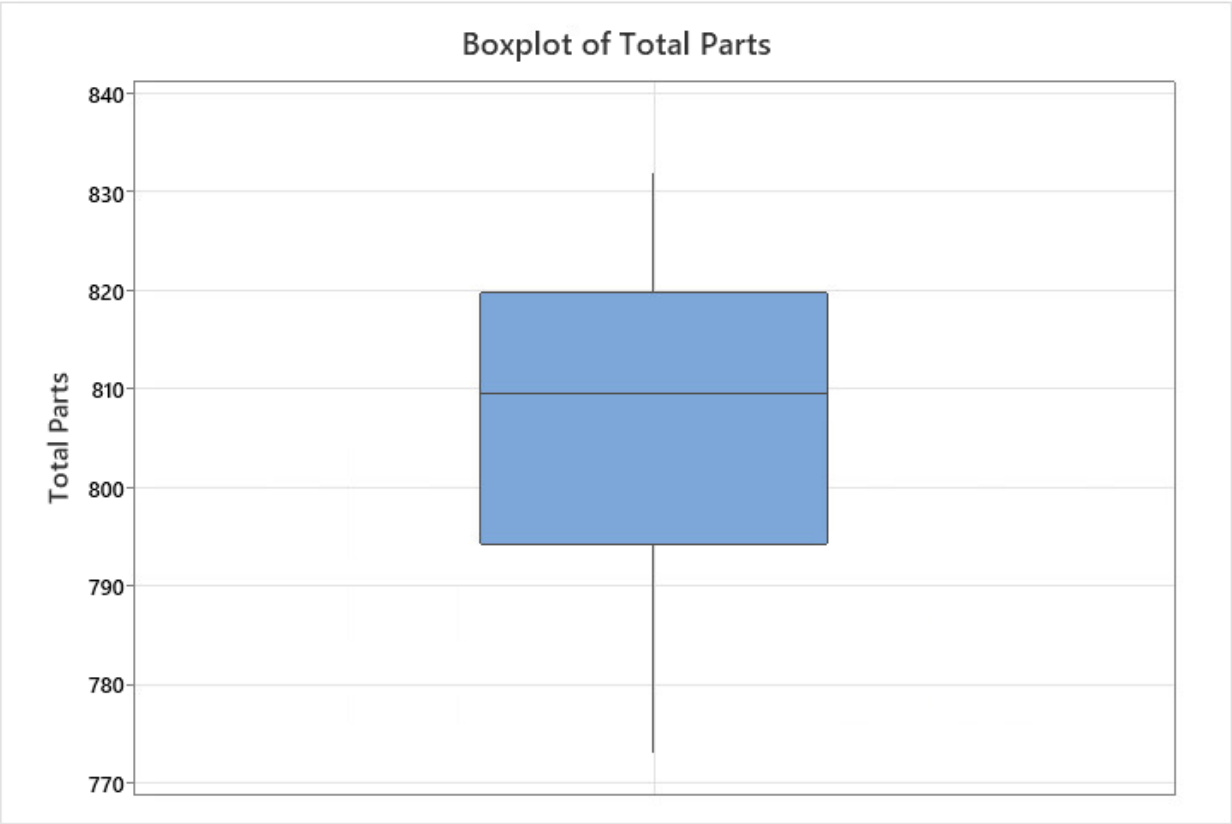

FIGURE 3. BOXPLOT OF TOTAL PARTS

The interquartile range extends from approximately 795 to 820 parts, indicating a relatively tight clustering of production levels around the median. The median lies near the center of the box, suggesting balanced variation above and below the midpoint. The whiskers reach the minimum and maximum observed values without identifying any outliers, consistent with stable production performance.

Figure 4 shows a probability plot of the daily total parts to assess whether the data follow a normal distribution.

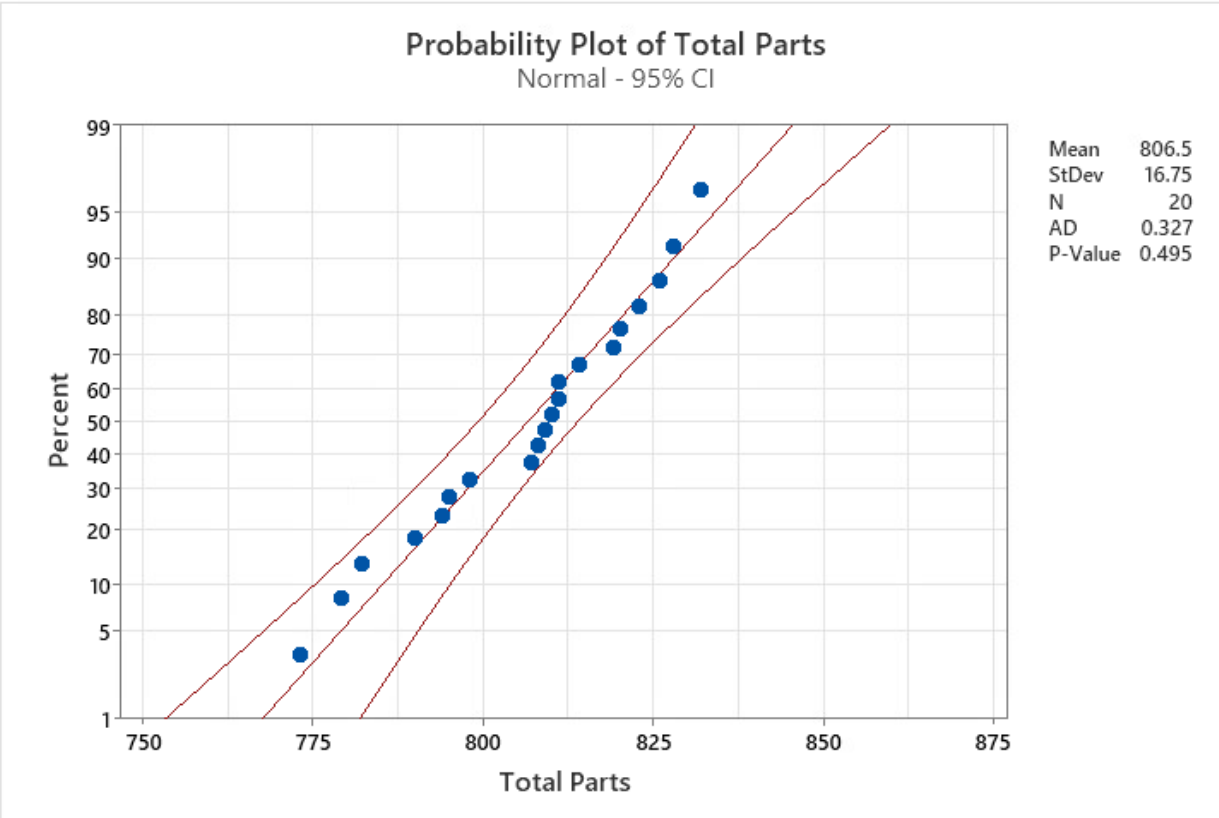

FIGURE 4. PROBABILITY PLOT OF TOTAL PARTS

The points follow the reference line closely and remain within the confidence bands, and the Anderson–Darling p-value of 0.495 provides no evidence to reject normality. This supports the impression that total daily production follows a reasonably normal distribution with no heavy tails or irregularities. The agreement across graphical displays reinforces the conclusion that production volume behaves consistently across days.

5.1.2. Numerical Summary

Table 1 presents the key descriptive statistics used to characterize the daily total part counts.

TABLE 1. DESCRIPTIVE STATISTICS OF TOTAL PARTS

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Total Parts	20	0	806.45	3.75	16.75	773.00	794.25	809.50	819.75	832.00

Across the 20 days of data, the mean daily production level was 806.45 parts with a standard deviation of 16.75, indicating moderate variation in total output from day to day. The middle 50% of observations ranged from 794.25 to 819.75 parts, showing that most production days fell within a

relatively narrow band around the center. The minimum and maximum values were 773 and 832 parts, both within a reasonable distance from the core distribution. These statistics collectively indicate that overall production volume remained consistent across the 20-day period. Through ***Equation 1***, the 95% confidence interval is (798.61, 814.29).

Figure 5 presents the I–MR chart, which is used to evaluate whether the process mean and variation are statistically stable over time.

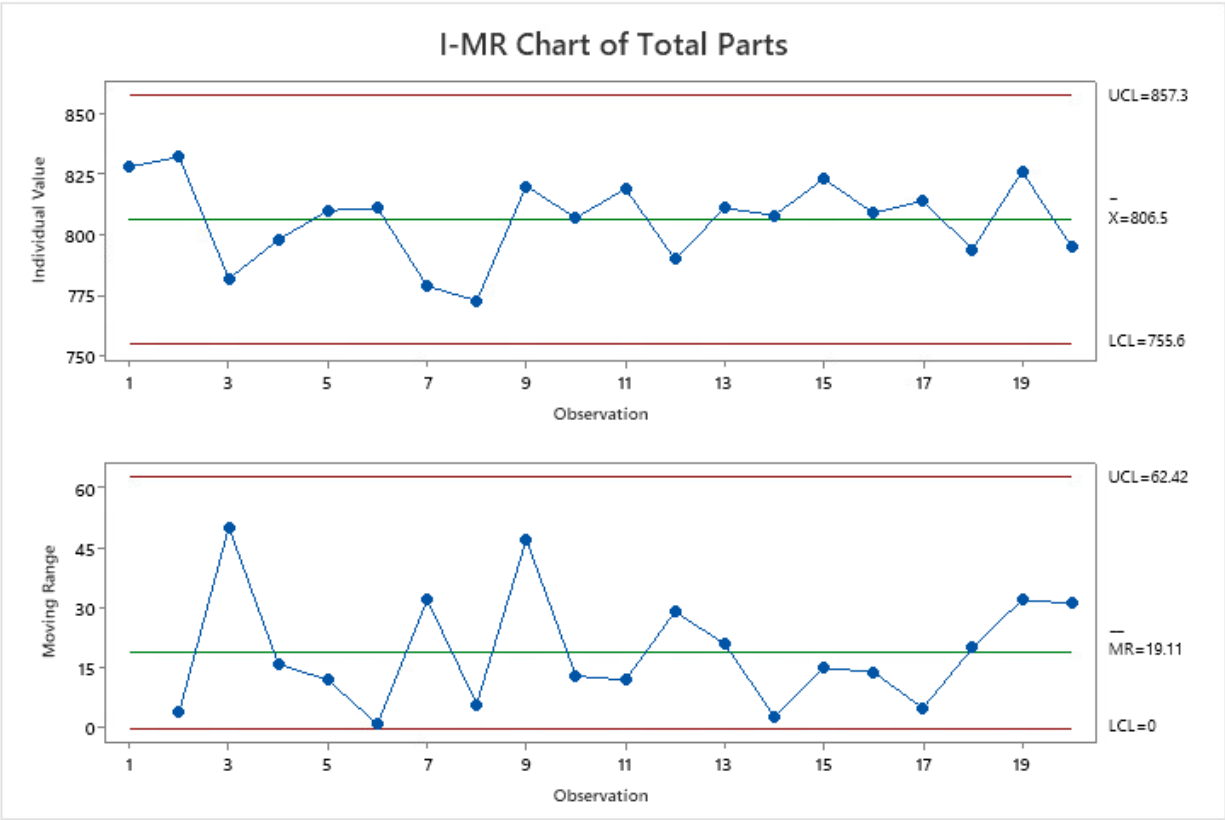

FIGURE 5. I-MR CHART OF TOTAL PARTS

The Moving Range chart is reviewed first to evaluate the stability of day-to-day changes in total output. All moving ranges fall within the control limits, indicating that the short-term variation in total production was stable and free of unusually large shifts. With stable variation, the control limits on the Individuals chart can be interpreted meaningfully. The Individuals chart shows all observations within their respective control limits, with no evidence of shifts, trends, or unusual patterns. These results demonstrate that the overall production volume behaved consistently from day to day, and the descriptive statistics above provide a reliable representation of the long-run performance of the total output process.

5.1.3. Predictions of Future Performance

Using the statistics from the numerical summary with ***Equation 2***, the total number of parts produced in a day is expected to fall between 770 and 843 with 95% confidence. The exact interval calculated was (770.53, 842.37).

5.2. Benchmark for Good Parts

5.2.1. Graphical Analysis

Figure 7 shows a run chart of the daily good parts to evaluate whether the process exhibits non-random patterns over time.

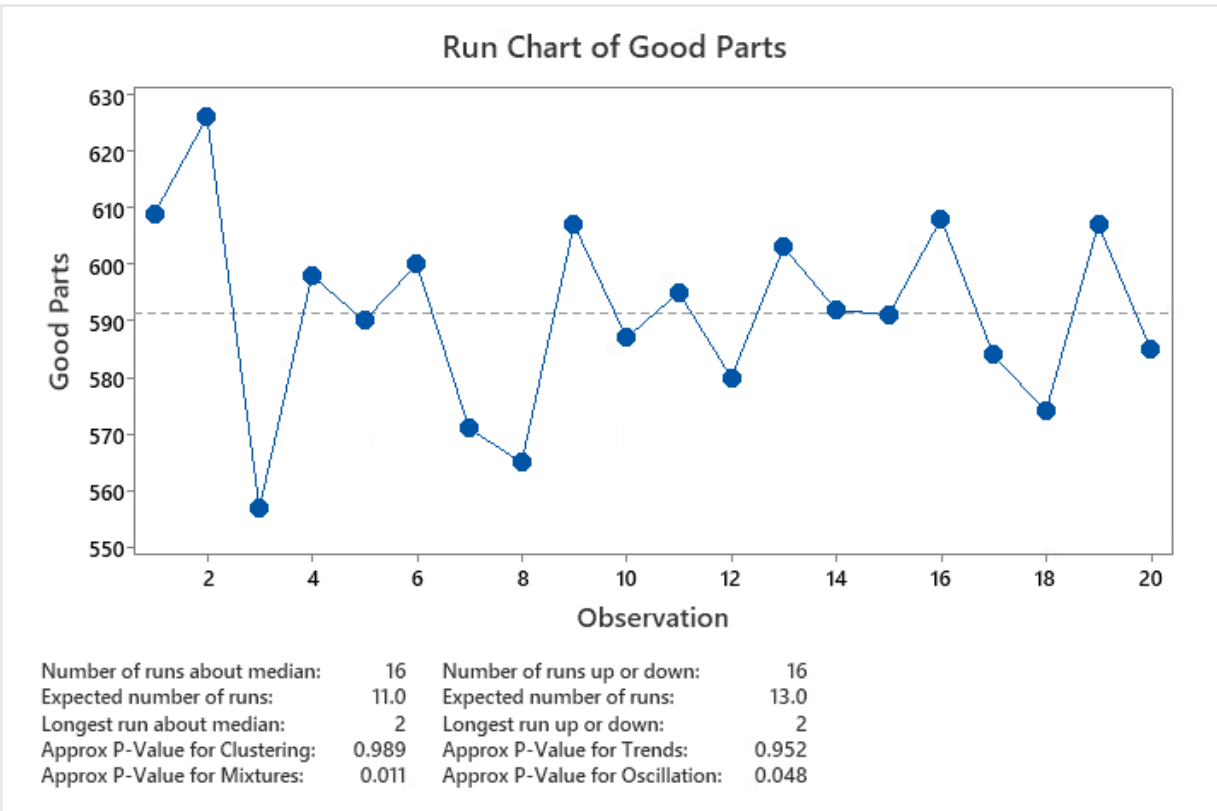

FIGURE 6. RUN CHART OF GOOD PARTS

The observations fluctuate around the median without extended runs, shifts, or trends. The number of runs is close to the expected value under randomness, and the p-values for clustering and trends do not indicate non-random structure. Although the mixture p-value is low, the plot does not show a meaningful separation or alternating pattern. Overall, the run chart suggests that the production of good parts remained steady throughout the 20-day period.

Figure 7 shows a histogram of the daily good parts to illustrate how the observations are distributed across their range of values.

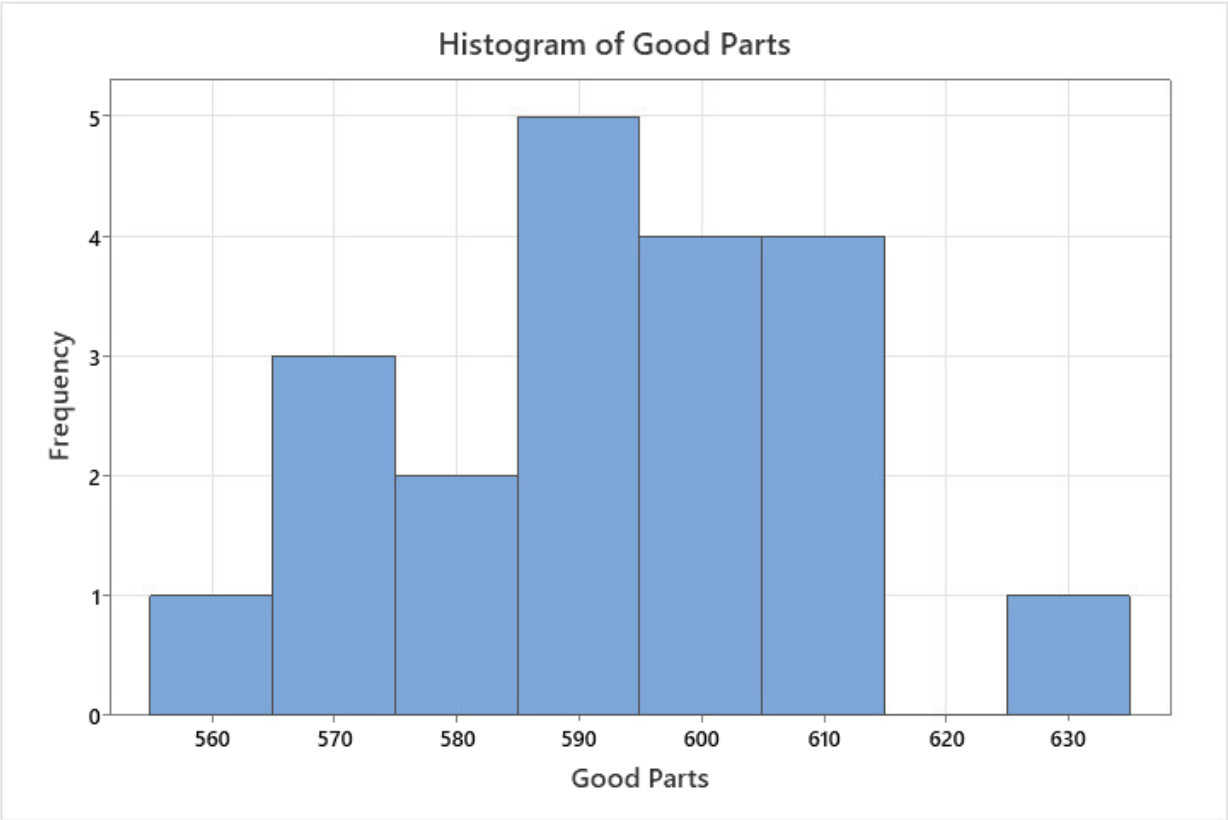

FIGURE 7. HISTOGRAM OF TOTAL PARTS

The distribution appears roughly symmetric, with most days producing between approximately 575 and 610 good parts. There are no extreme values or gaps, and the overall shape suggests a fairly regular pattern of daily output. The spread of the histogram indicates moderate variation, but no unusually high or low production days that would suggest instability.

Figure 8 shows a boxplot of the daily good parts to summarize the central tendency and variability of the data.

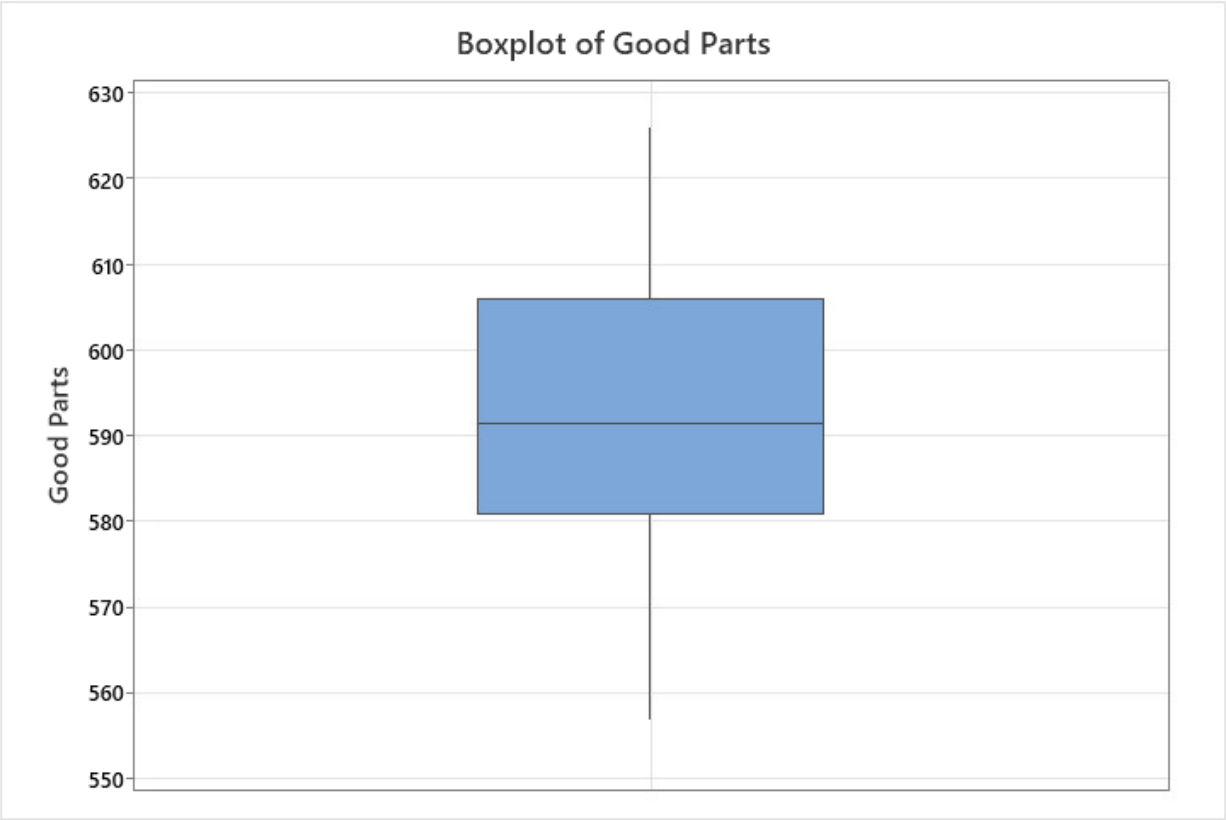

FIGURE 8. BOX PLOT OF GOOD PARTS

The interquartile range extends from roughly 582 to 605 parts, indicating a tight central cluster of daily output. The median lies near the center of the box, suggesting balanced variation above and below middle production levels. The whiskers reach the minimum and maximum values without identifying any outliers, meaning that all observed days fall within a reasonable range of expected production performance.

Figure 9 shows a probability plot of the daily good parts to assess whether the data follow a normal distribution.

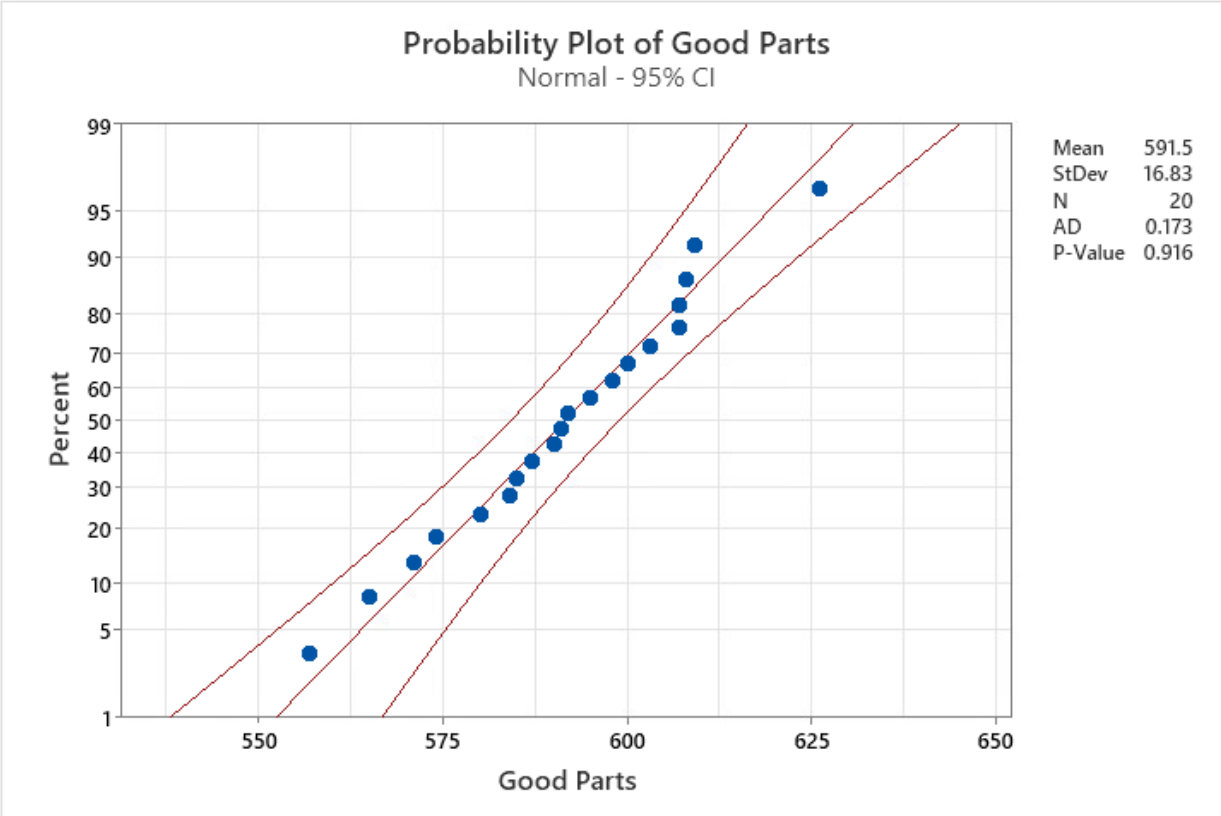

FIGURE 9. PROBABILITY PLOT OF GOOD PARTS

The points closely follow the reference line within the confidence bands, and the Anderson–Darling p-value of 0.916 provides no evidence against normality. This suggests that the distribution of good-part output is consistent with a normal model, reinforcing the interpretation that daily production levels fluctuate in a statistically regular manner without heavy-tailed or irregular behavior.

5.2.2. Numerical Summary

Table 2 presents the key descriptive statistics used to characterize the daily good part counts.

TABLE 2. DESCRIPTIVE STATISTICS OF GOOD PARTS

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Good Parts	20	0	591.45	3.76	16.83	557.00	581.00	591.50	606.00	626.00

Across the 20 days of data, the mean number of good parts produced per day was 591.45 with a standard deviation of 16.83, reflecting moderate day-to-day variation in successfully produced units. The middle 50% of observations ranged from 581 to 606 parts, indicating that the majority of production days fell within a relatively narrow band around the center. The minimum and maximum values, 557 and 626, fall within a reasonable distance of the mean and appear consistent with the

natural spread of the process rather than unusually high or low production outcomes. These statistics collectively indicate that the process maintained a stable level of successful output during the observed period. Through ***Equation 1***, the 95% confidence interval is (583.57, 599.33).

Figure 10 presents the I–MR chart, which is used to evaluate whether the process mean and variation are statistically stable over time.

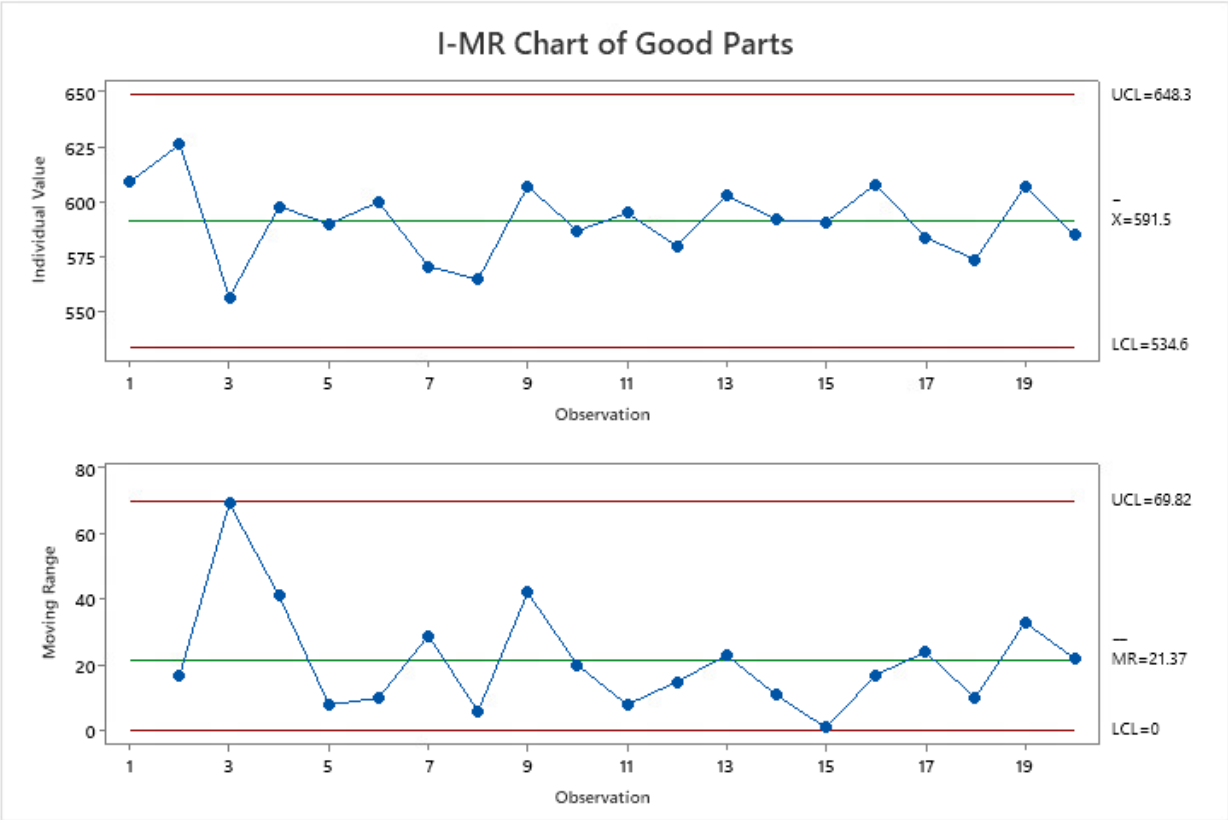

FIGURE 10. I-MR CHART OF GOOD PARTS

The Moving Range chart is examined first to assess the stability of day-to-day changes in the number of good parts produced. All moving-range values fall within the control limits, indicating that the short-term variation was stable and free of unusual spikes. With stable variation, the control limits on the Individuals chart can be interpreted reliably. All individual observations fall within their control limits, and no shifts, trends, or extended runs appear in the data. These results suggest that the mean number of good parts produced per day remained statistically consistent throughout the 20-day period, and the descriptive statistics above provide a valid representation of long-run performance for this output category.

5.2.3. Predictions of Future Performance

Using the statistics from the numerical summary with ***Equation 2***, the number of good parts produced in a day is expected to fall between 555 and 628 with 95% confidence. The exact interval calculated was (555.35, 627.55).

5.3. Check Sheet and Pareto Chart Analysis

Using the quality data from the Mouse Factory data center, a check sheet (***Table 5***) and a corresponding Pareto chart were created to analyze the weight of certain defects amongst the population of both bad and out of speculation parts, shown in ***Figure 11***.

TABLE 3. CHECK SHEET OF QUALITY NONCONFORMITIES

NONCONFORMITY	WEEK 1	WEEK 2	WEEK 3	WEEK 4	TOTAL
Base-Black Streaking	5	5	3	4	17
Base-Bubble	C	2	5	5	16
Base-Burn Spot	2	5	3	3	13
Base-Mold Flashing	5	5	6	5	21
Base-Part Stuck in Mold	88	83	99	89	359
Base-Poor Weld Line	5	7	1	3	16
Base-Short Shot	3	2	2	5	12
Base-Sink Spot		5	2	5	12
Base-Snap Point Diameter	9	11	10	12	42
Base-Snap Point Opening	14	15	10	12	51
Base-Surface Imperfection	2	7	4	3	16
Base-Warpage	1	4	3	5	13
Button-Black Streaking	2	5	6	6	19
Button-Bubble	4	2	2	4	12
Button-Burn Spot	3	6	4	4	17
Button-Hole Diameter	263	255	269	285	1072
Button-Mold Flashing	72	68	92	67	299
Button-Part Stuck in Mold	1	4	5	6	16
Button-Poor Weld Line	4	4	5	6	19
Button-Short Shot	2	1	2	2	7
Button-Sink Spot	2	4	4	8	18
Button-Surface Imperfection	7	5	2	2	16
Button-Warpage	5	2	7	2	16
Cover-Black Streaking	6	3	3	4	16
Cover-Bubble	8		2	1	11
Cover-Burn Spot	7	4	2	5	18
Cover-Mold Flashing	4	1	6	5	16
Cover-Part Stuck in Mold	2	2	2	4	10

Cover-Poor Weld Line	3	4	6	2	15
Cover-Shaft Diameter	10	11	7	10	38
Cover-Short Shot	4	4	1	4	13
Cover-Sink Spot	2	3	3		8
Cover-Surface Imperfection	8	8	6	2	24
Cover-Warpage	5	4	4	4	17
Diffuser-Black Specks	8	6	2	3	19
Diffuser-Brittleness	3	9	2	5	19
Diffuser-Burn Marks	5	4	5	3	17
Diffuser-Discoloration	3	6	3	10	22
Diffuser-Intensity	12	12	9	12	45
Diffuser-Jetting	1	2	6	2	11
Diffuser-Part Stuck in Mold	196	203	228	208	835
Diffuser-Splay Marks	7	5	5	2	19
Diffuser-Voids	1	2	4	8	15
Diffuser-Warpage	2	5	7	6	20
Diffuser-Weld lines	4	2	3	1	10
Scroll Base-Black Streaking	2	5	5	5	17
Scroll Base-Bubble	3	3	2	3	11
Scroll Base-Burn Spot	4	3	4	2	13
Scroll Base-Hole Diameter	12	4	8	8	32
Scroll Base-Mold Flashing	6	4	5	3	18
Scroll Base-Opening	9	15	13	13	50
Scroll Base-Part Stuck in Mold	4	3	3	3	13
Scroll Base-Pin Diameter	11	9	14	13	47
Scroll Base-Poor Weld Line	2	2	4	7	15
Scroll Base-Short Shot	4	7	2	1	14
Scroll Base-Sink Spot	3	5	4	3	15
Scroll Base-Surface Imperfection	4	2	3	4	13
Scroll Base-Warpage	5	4		3	12
Scroll Wheel Assembly-Black Streaking	4	1	3	3	11
Scroll Wheel Assembly-Bubbles	4	4	9	3	20
Scroll Wheel Assembly-Burn Spot	4	3	6	3	16
Scroll Wheel Assembly-Missing Teeth	218	209	177	201	805
Scroll Wheel Assembly-Mold Flashing	3	2	3	4	12
Scroll Wheel Assembly-Part Stuck in Mold	3	9	2	2	16
Scroll Wheel Assembly-Shaft Pin Diameter	85	97	107	106	395
Scroll Wheel Assembly-Short Shot	4	2	5	4	15
Scroll Wheel Assembly-Sink Spot	7	5	2	5	19
Scroll Wheel Assembly-Surface Imperfection	1	8	5	4	18
Scroll Wheel Assembly-Warpage	3	4	4	4	15
(blank)					
Grand Total	1209	1217	1252	1251	4929

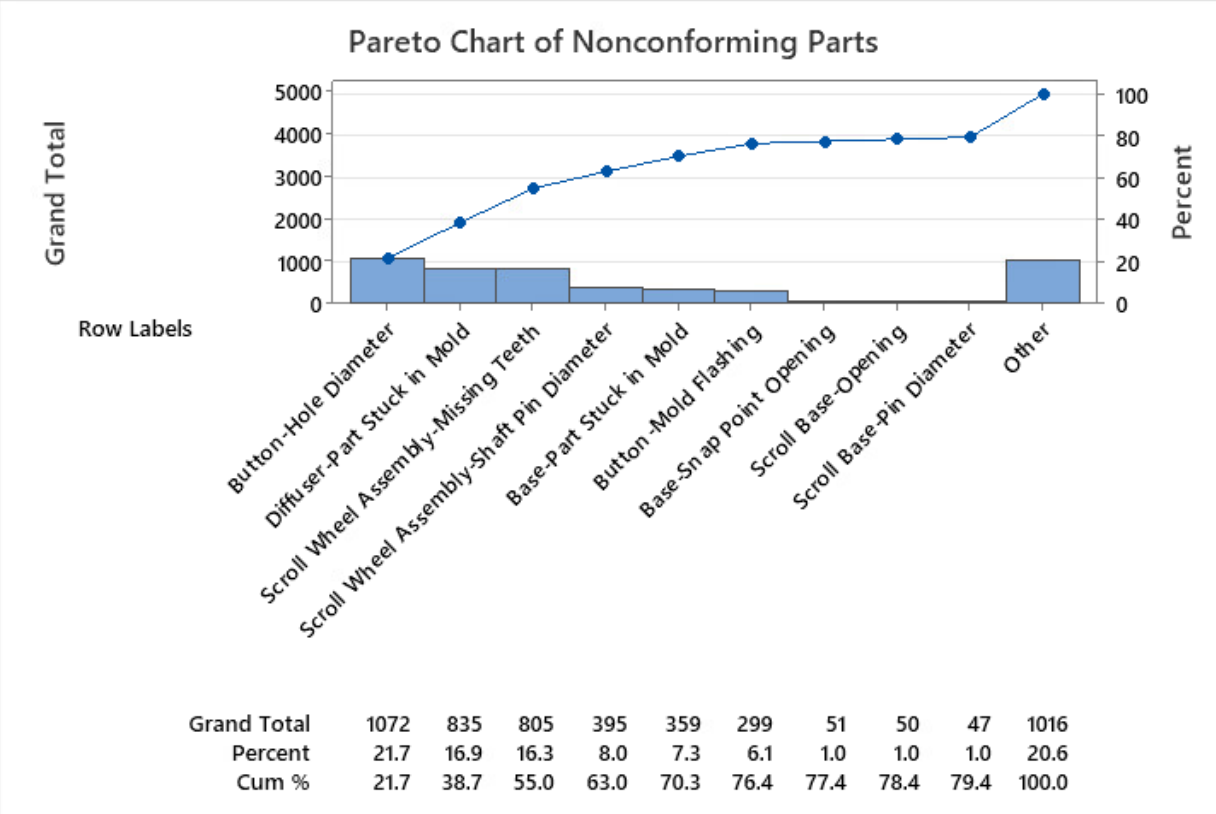

FIGURE 11. PARETO CHART OF NONCONFORMING PARTS

Out of the 69 types of nonconformities listed on the check sheet, the Pareto chart shows that 9 types account for 79.4% of all nonconforming parts, with the remaining 20.6% grouped into the "Other" category. Furthermore, 6 types account for 76.3%, and 3 types account for 54.9%. All nonconformity types included in "Other" represent less than 1% each of the total. Each nonconformity type is classified as either an off-spec part or a bad part.

5.4. Benchmark of Off-spec Parts

Figure 11 shows that Button-Hole Diameter (21.7%), Scroll Wheel Assembly–Shaft Pin Diameter (8%), Scroll Base-Opening (1%), and Scroll Base-Pin Diameter (1%) nonconformities fall under the classification of off-spec parts. Combined, these categories account for 31.7% of all observed nonconformities without considering those grouped in "Other". Because off-spec parts represent dimensional violations rather than functional failures, this benchmark evaluates the typical daily level of off-spec production to assess the stability and long-run behavior of this portion of the process.

5.4.1. Graphical Analysis

Figure12 provides a run chart to examine how off-spec counts behave over time and to detect any non-random temporal patterns.

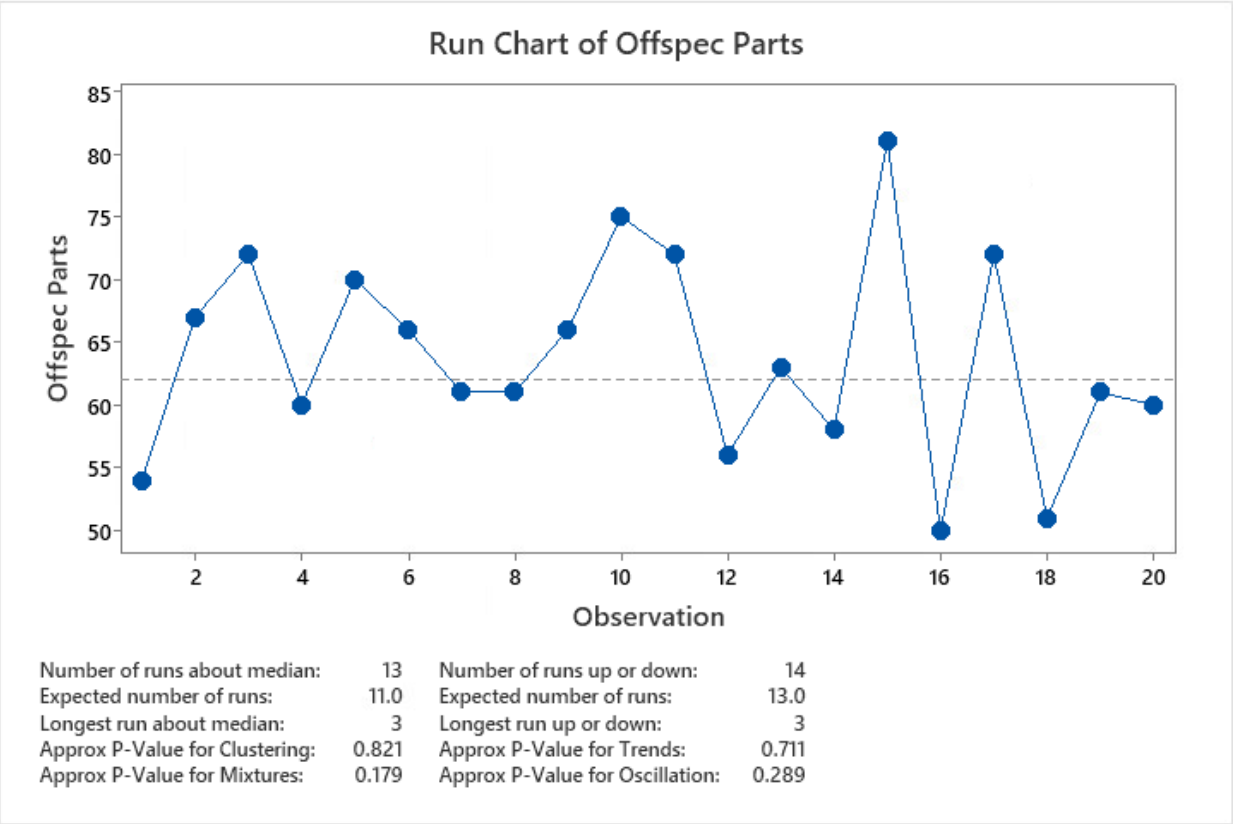

FIGURE 12. RUN CHART OF OFF-SPEC PARTS

The run chart highlights the sequence and timing of off-spec values, showing how the process behaves from day to day. The counts oscillate within a relatively tight range without forming long stretches of increasing, decreasing, or consistently high/low values. The statistical run tests support this visual impression: the number of runs, the longest run lengths, and the p-values for trend, oscillation, clustering, and mixtures all meet expected conditions. These results indicate that the data fluctuates randomly over time rather than showing systematic temporal shifts or cycles. Although the run chart does not confirm statistical control, it supports the view that the process is not influenced by major day-to-day disturbances.

Figure 13 shows a histogram of the daily off-spec parts to illustrate how the observations are distributed across their range of values.

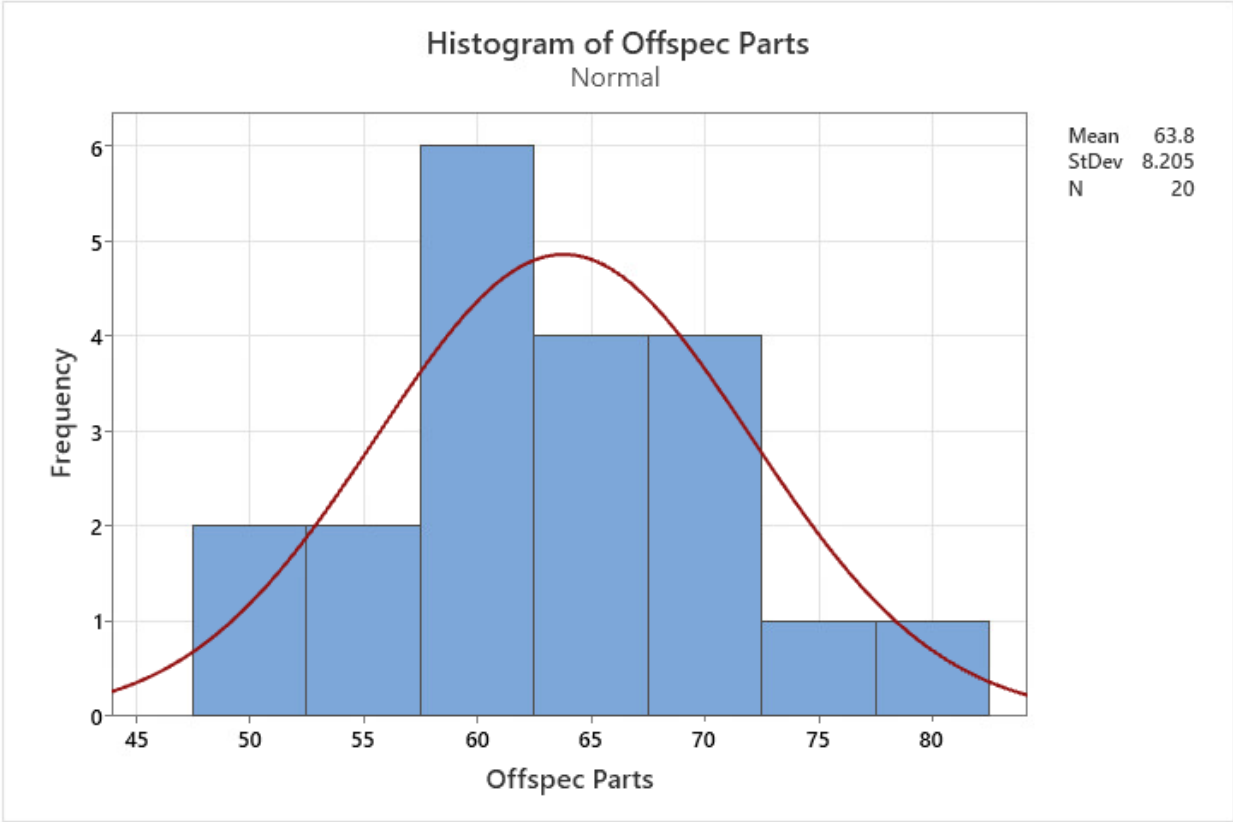

FIGURE 13. HISTOGRAM OF OFF-SPEC PARTS

The histogram reveals how frequently different off-spec part counts occur. The distribution is unimodal and reasonably symmetric, with most observations occurring between about 55 and 70 parts. The counts do not cluster at extreme values, and no gaps or unusual clumps appear. The smooth normal curve overlays the bars closely, indicating that the data's overall shape is

consistent with a bell-shaped distribution. This graphical structure suggests a process where typical daily output does not differ drastically from one day to the next.

Figure 14 displays a boxplot summarizing the central tendency, spread, and potential outliers of the off-spec part counts.

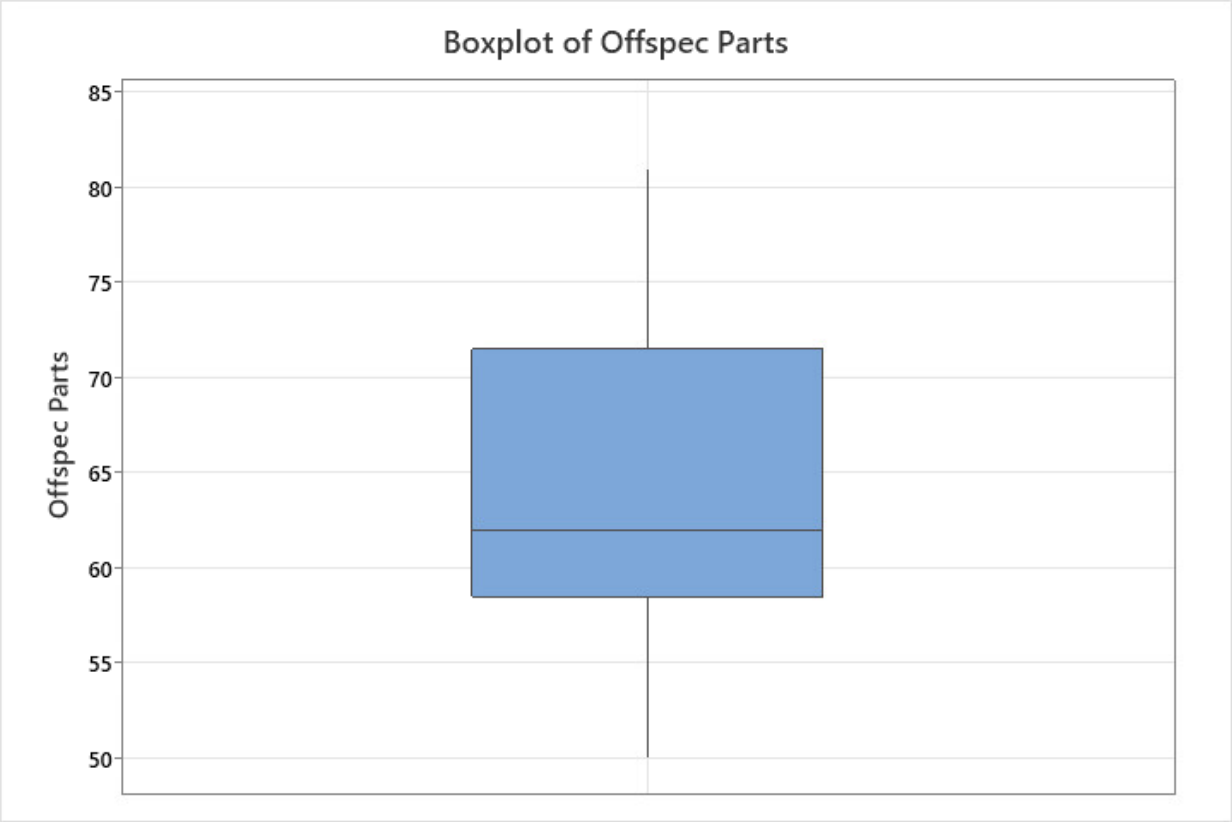

FIGURE 14. BOX PLOT OF OFF-SPEC PARTS

The boxplot emphasizes the variability of the off-spec counts in a compact form. The median is positioned near the middle of the interquartile range, indicating balanced variation above and below the center of the data. The IQR is relatively narrow, meaning that half of all observations fall within a tight band around the median. The whiskers extend to the lowest and highest daily values without identifying outliers, suggesting that the extremes of the dataset are still typical and not the

result of rare or unusual events. This visualization highlights that the process displays controlled and moderate variability.

Figure 15 presents a probability plot used to formally assess whether the off-spec counts follow a normal distribution.

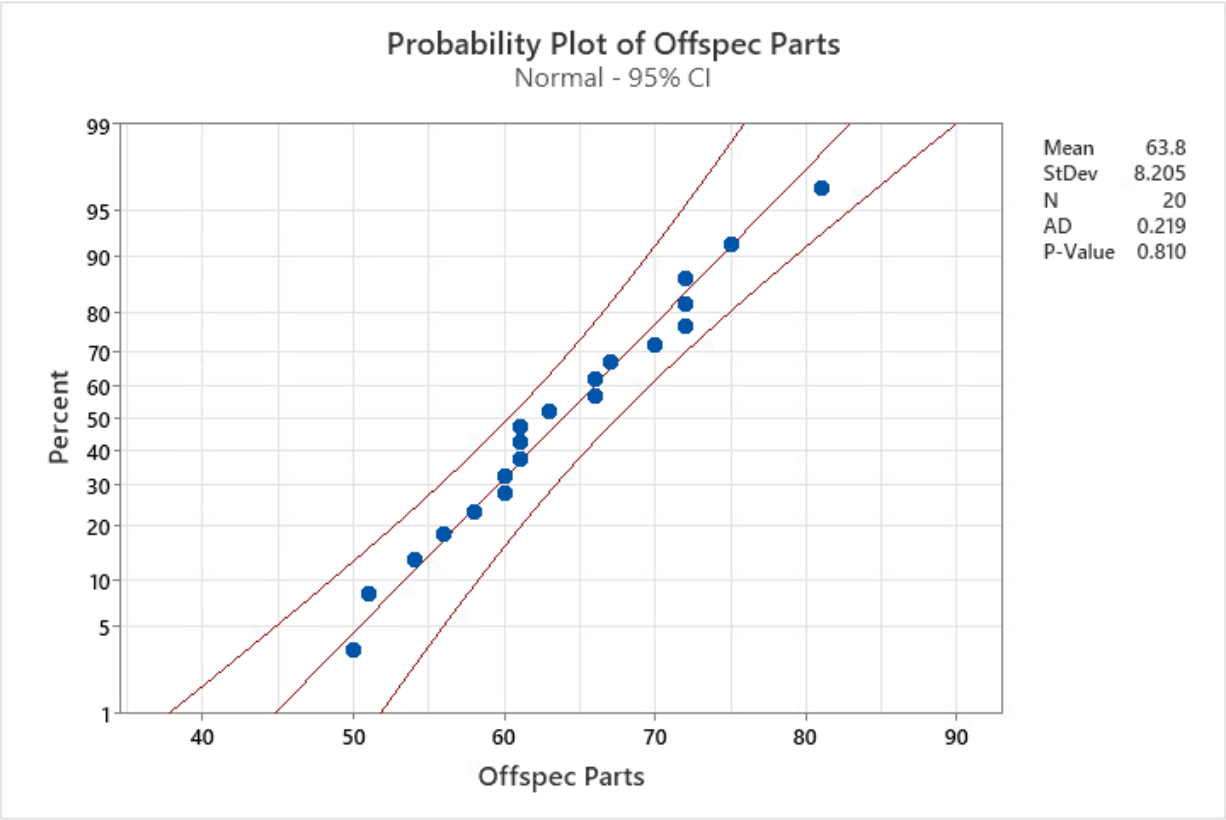

FIGURE 15. PROBABILITY PLOT OF OFF-SPEC PARTS

The probability plot focuses on how well the data align with theoretical normal behavior. Nearly all points fall close to the reference line, and the Anderson–Darling statistic (0.219) is small, indicating an excellent overall fit. The p-value of 0.810 provides strong evidence that the data are consistent with normality, meaning there is no statistical reason to suspect skewness or heavy tails. Because the observed values fall comfortably within the 95% confidence bands, the normal model provides an appropriate basis for constructing confidence intervals and using the sample mean for benchmark estimation.

5.4.2. Numerical Summary

Table 4 presents the key descriptive statistics used to characterize the daily off-spec part counts.

TABLE 4. DESCRIPTIVE STATISTICS OF OFF-SPEC PARTS

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Offspec Parts	20	0	63.80	1.83	8.21	50.00	58.50	62.00	71.50	81.00

The mean number of off-spec parts was 63.80, and the standard deviation was 8.21, indicating moderate day-to-day variation in the process. Through ***Equation 1***, the 95% confidence interval is (59.96, 67.64). This interval provides a statistically supported range for the long-run average level of off-spec production, and the roughly symmetric distribution observed earlier supports the appropriateness of these summary measures.

Figure 16 presents the I–MR chart, which is used to evaluate whether the process mean and variation are statistically stable over time.

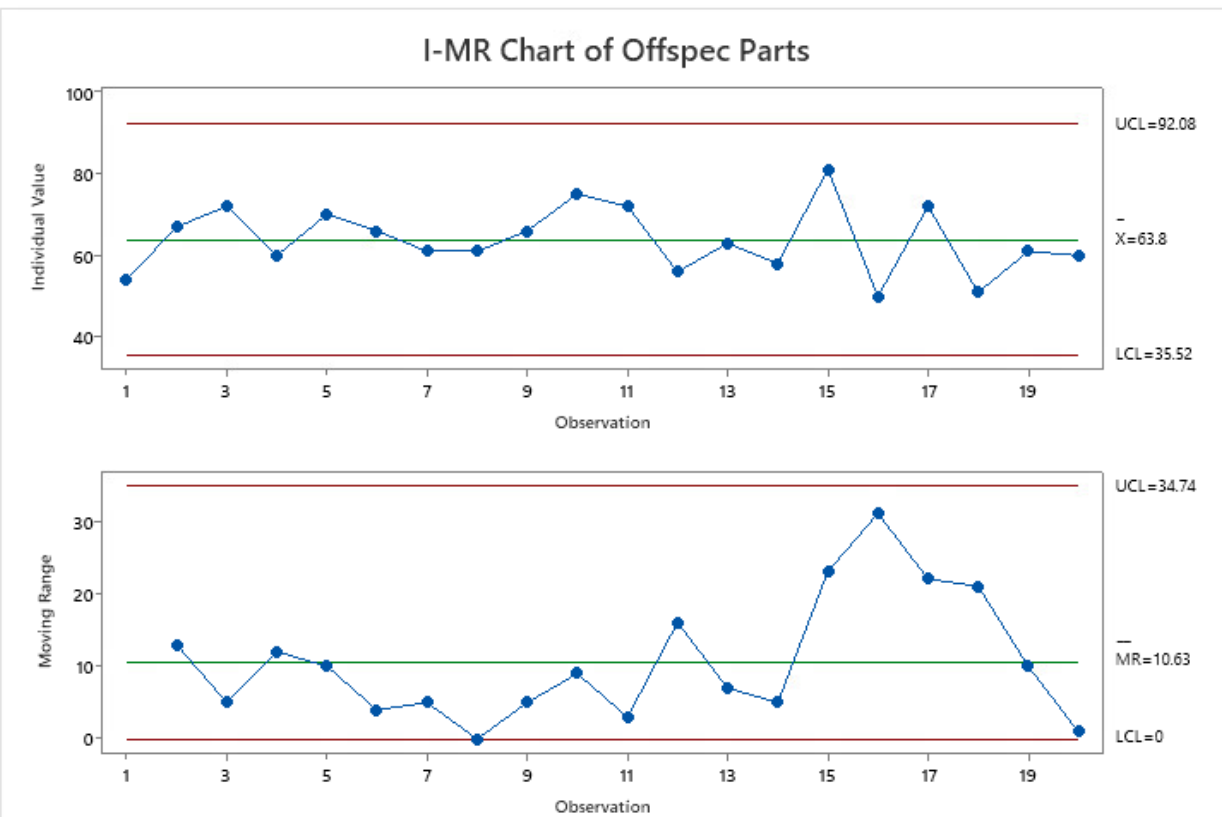

FIGURE 16. I-MR CHART OF OFF-SPEC PARTS

The Moving Range chart shows no points outside of the control limits, indicating that the day-to-day variation in off-spec output is consistent with expected natural variability. Because the MR chart is stable, the control limits on the Individuals chart are reliable for assessing the mean. The Individuals chart also displays no points outside the control limits and no sustained patterns that would signal trends, shifts, or other non-random behavior. Together, these results demonstrate that both the mean and the variation of the process are statistically stable, allowing the sample mean and standard deviation to be used confidently for prediction and confidence interval estimation.

5.4.3. Predictions of Future Performance

Using the statistics from the numerical summary with ***Equation 2***, the number of good parts produced in a day is expected to fall between 50 and 86 with 95% confidence. The exact interval calculated was (50.91, 85.69).

5.5. Benchmark of Bad Parts

Figure 11 shows that Diffuser-Part Stuck in Mold (16.9%), Scroll Wheel Assembly-Missing Teeth (16.3%), Base Part Stuck in Mold (7%), Button-Mold Flashing (6.1%), and Base-Snap Point Opening (1%) nonconformities fall under the classification of bad parts. Combined, these nonconformities account for 68.2% of all observed nonconformities without considering those grouped in "Other". Because bad parts represent functional failures, this benchmark evaluates the typical daily level of bad production to assess the stability and long-run behavior of this portion of the process.

5.5.1. Graphical Analysis

Figure 17 shows a run chart of the daily bad-part counts to assess potential time-related patterns such as shifts, trends, or cycles.

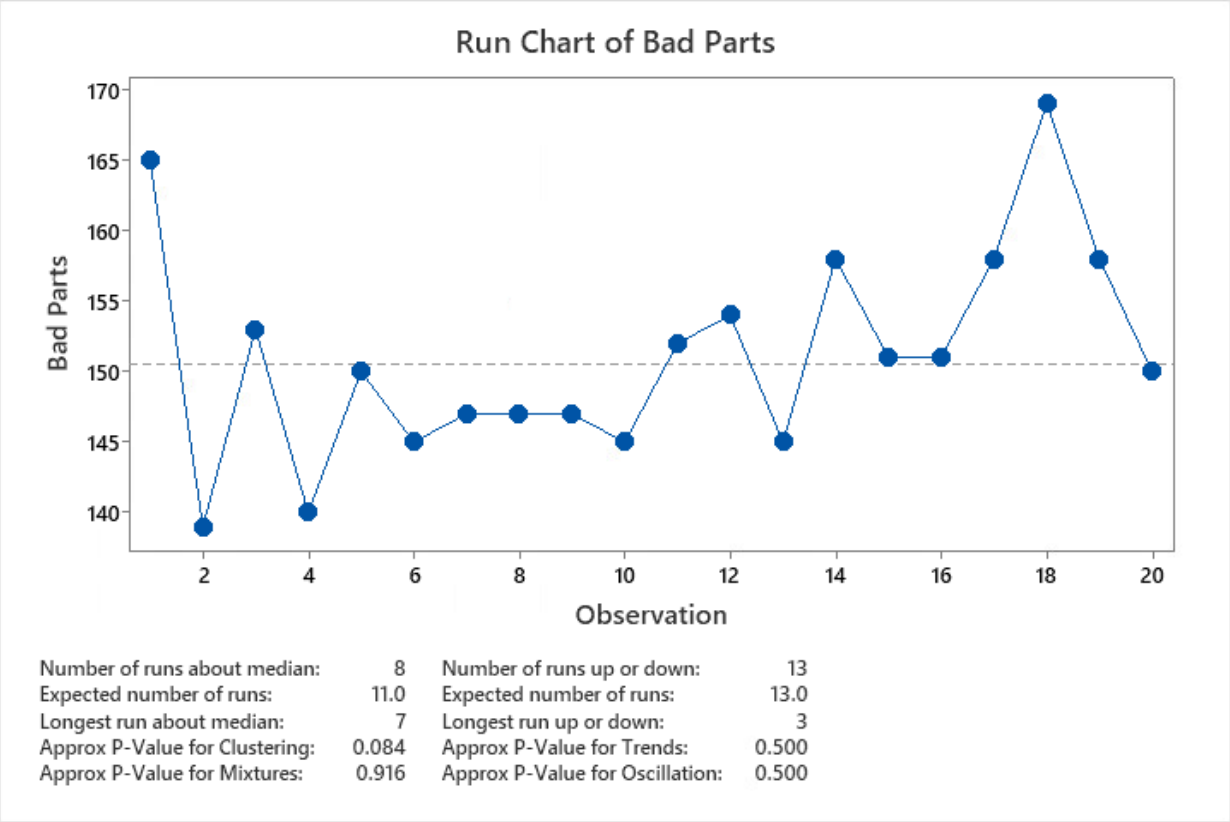

FIGURE 17. RUN CHART OF BAD PARTS

The run chart displays day-to-day fluctuations without sustained increases or decreases, and the values remain within a relatively consistent range. The run test results show p-values above 0.05 for clustering, mixtures, trends, and oscillation, indicating no statistically meaningful non-random patterns. The number of runs is close to the expected value, and the longest run lengths fall within normal limits. These results suggest that the daily number of bad parts varies randomly over time and does not exhibit systematic temporal behavior.

Figure 18 presents a histogram of the daily bad-part counts to illustrate the overall distribution of the process output.

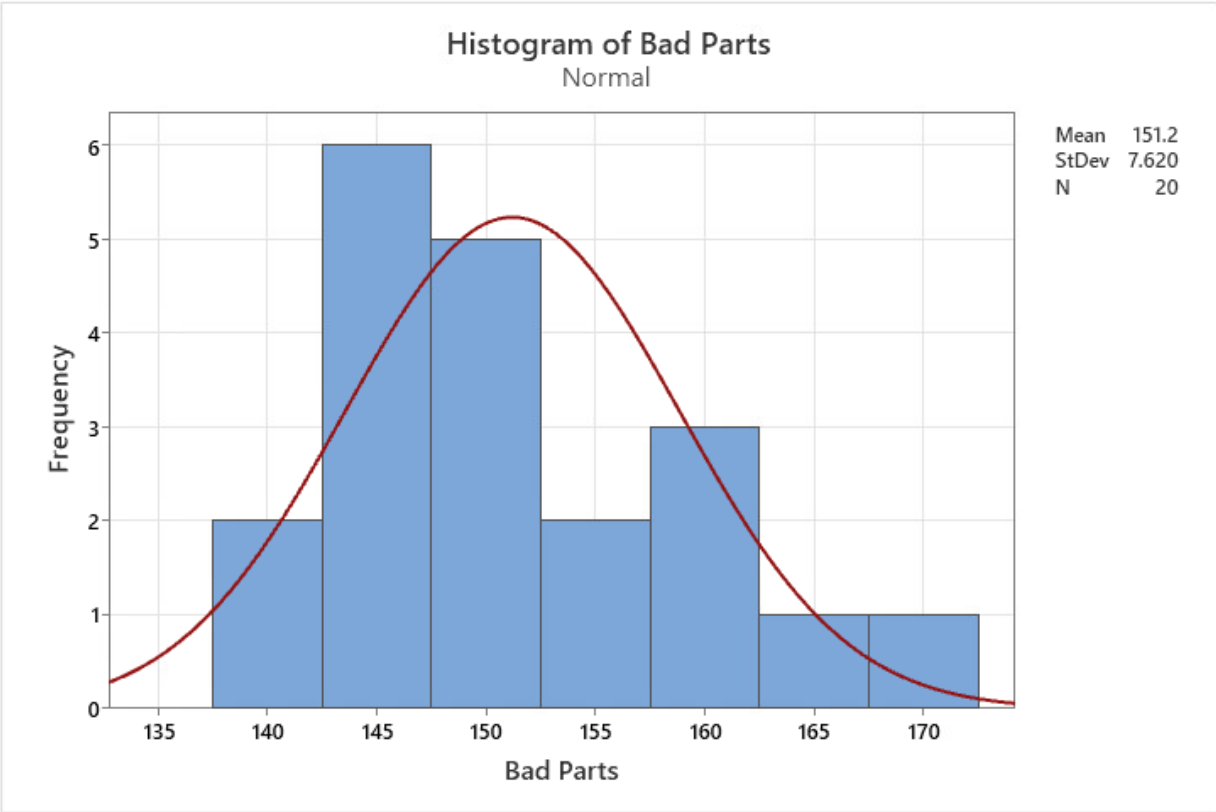

FIGURE 18. HISTOGRAM OF BAD PARTS

The histogram shows a unimodal distribution centered around roughly 150 bad parts per day. Most observations fall between approximately 140 and 160, and the distribution appears fairly symmetric. The overlaid normal curve aligns closely with the bar pattern, suggesting that the daily number of bad parts follows a pattern reasonably consistent with normal variation. There are no large gaps or extreme values, indicating that the day-to-day variation seen in the data is typical for this process.

Figure 19 displays a boxplot summarizing the central tendency and variability of the daily bad-part counts.

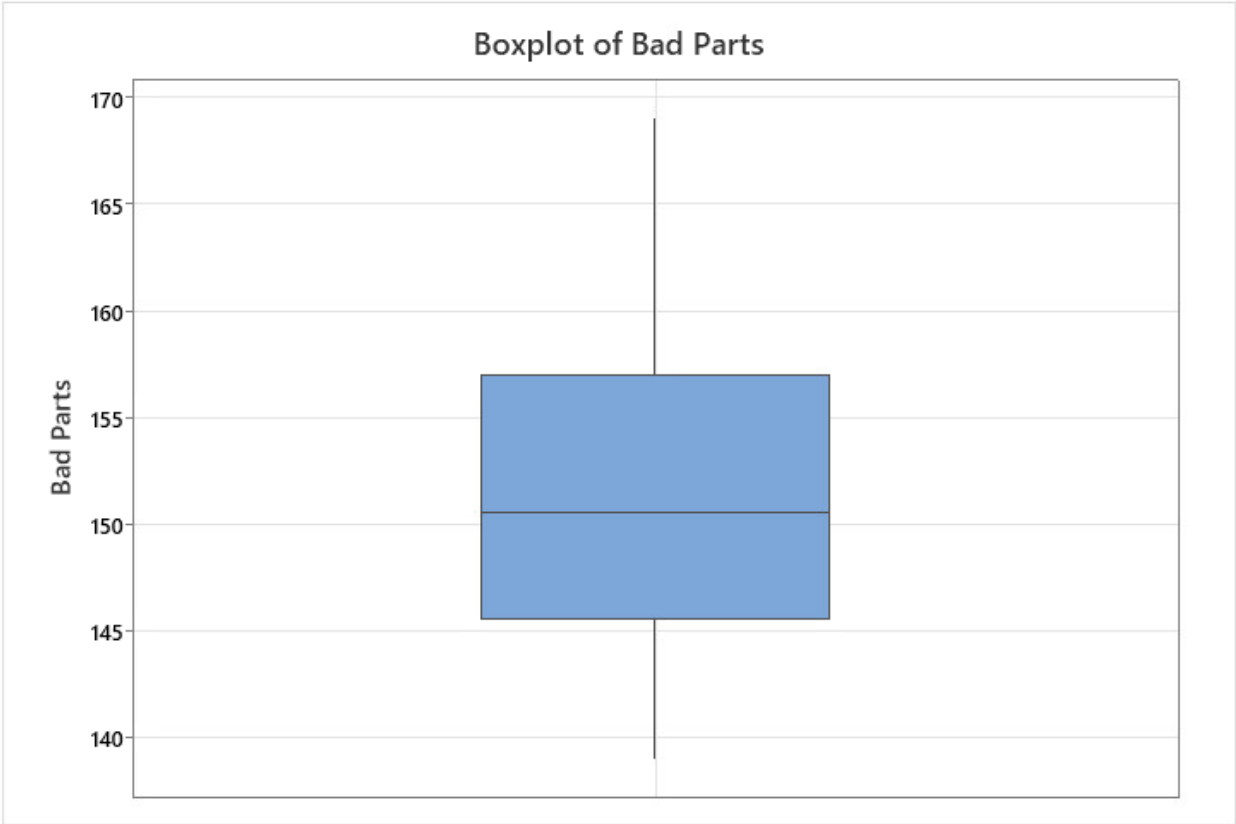

FIGURE 19. BOX PLOT OF BAD PARTS

The boxplot shows a median near 151 parts, with an interquartile range spanning approximately 146 to 156 parts. The whiskers extend to roughly 139 and 169, and no outliers are identified. The box and whiskers appear balanced, indicating limited skewness and reinforcing the impression of a stable and moderately varying process. The concentration of values within a relatively narrow range around the median suggests consistent daily performance.

Figure 20 provides a probability plot to evaluate whether the distribution of daily bad-part counts is consistent with a normal distribution.

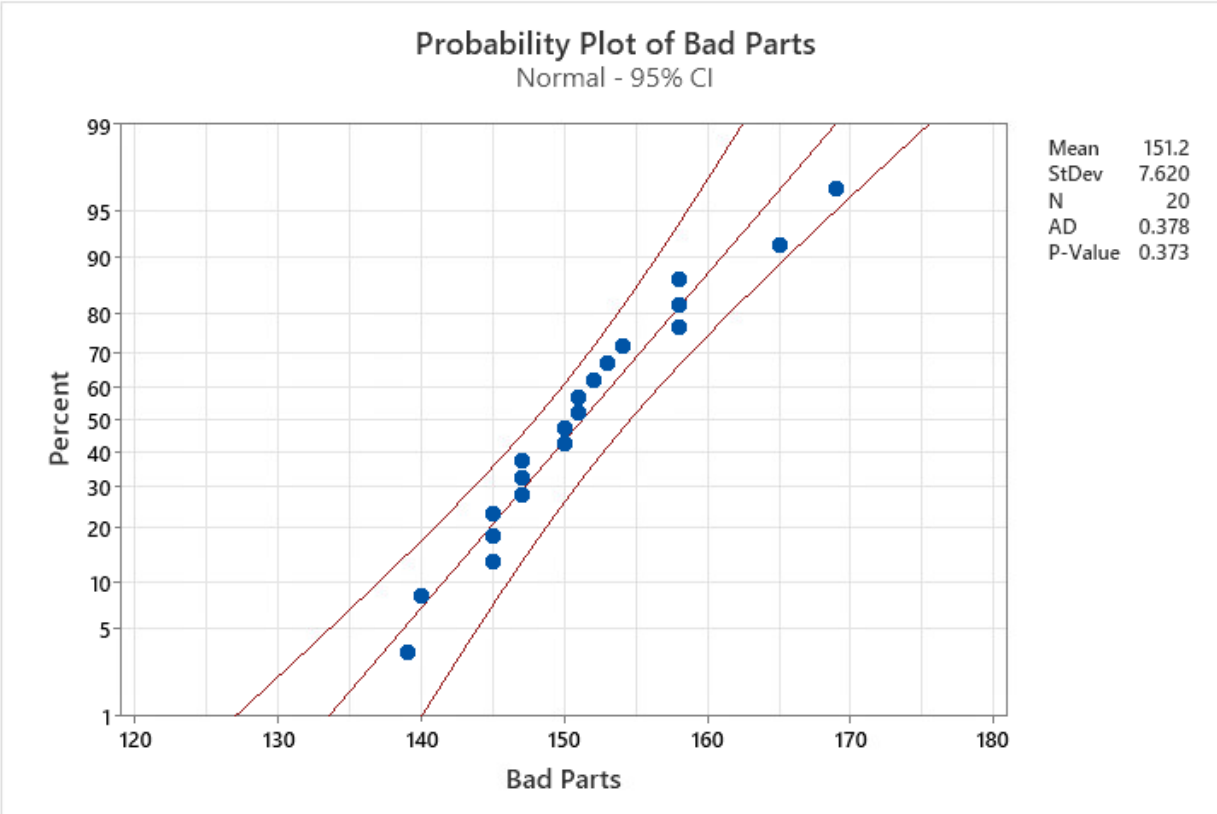

FIGURE 20. PROBABILITY PLOT OF BAD PARTS

Most points lie close to the reference line, and the data fall within the 95% confidence bands, indicating a good fit to the normal model. The Anderson–Darling statistic is low (AD = 0.378), and the p-value of 0.373 provides no evidence of non-normality. The lack of curvature or deviation from the reference line suggests that the normal distribution appropriately represents the process, supporting the use of common statistical summaries for this measure.

5.6. Numerical Summary

Table 5 presents the key descriptive statistics used to characterize the daily bad part counts.

TABLE 5. DESCRIPTIVE STATISTICS OF BAD PARTS

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Bad Parts	20	0	151.20	1.70	7.62	139.00	145.50	150.50	157.00	169.00

Across the 20 days of data, the mean daily number of bad parts was 151.2, with a standard deviation of 7.62, indicating a moderate level of day-to-day variation. The interquartile range extended from 145.5 to 157.0, showing that half of all observations fell within a relatively narrow band around the center of the distribution. The minimum and maximum values, 139 and 169, respectively, represent the natural spread of the process rather than unusually extreme days.

These statistics collectively describe a process with a stable central tendency and moderate, consistent variation across the observed period. Through ***Equation 1***, the 95% confidence interval is (147.63, 154.77).

Figure 21 presents the I–MR chart, which is used to evaluate whether the process mean and variation are statistically stable over time.

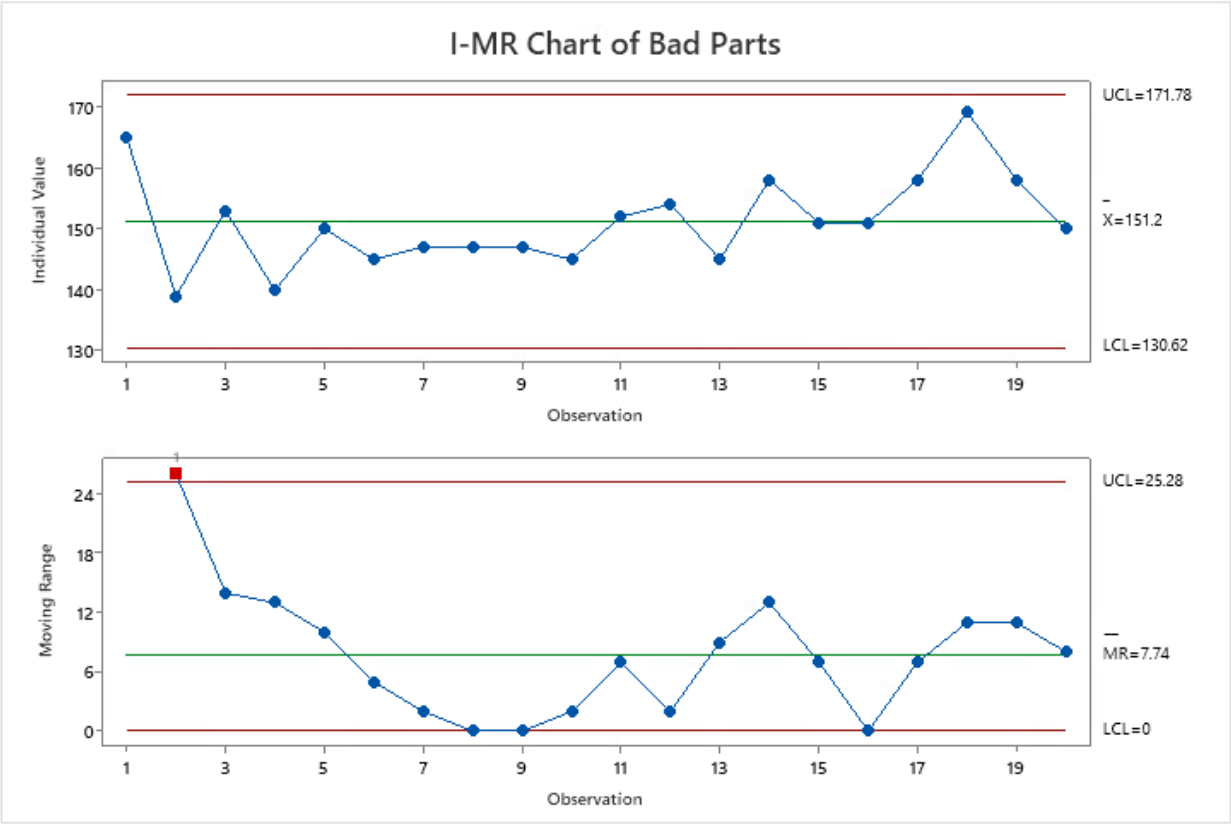

FIGURE 21. I-MR CHART OF BAD PARTS

The Moving Range chart is examined first to evaluate the stability of day-to-day variation. One moving-range value exceeds its upper control limit, indicating a single instance of unusually large change from one day to the next. Aside from this point, the remaining moving ranges fall within expected limits, suggesting generally stable short-term variation. Because the MR chart is mostly stable, the control limits on the Individuals chart are appropriate for assessing the behavior of the process mean. The Individuals chart shows no points outside the control limits and no indications of shifts, trends, or sustained runs. These results suggest that despite one isolated spike in variation, the process mean remained statistically consistent over the 20-day period, and the descriptive statistics above provide a valid representation of long-run bad-part performance.

5.6.1. Predictions of Future Performance

Using the statistics from the numerical summary with ***Equation 2***, the number of good parts produced in a day is expected to fall between 134 and 168 with 95% confidence. The exact interval calculated was (134.87, 167.53).

6. Conclusions

Although there is some indication of special cause variation for bad part production in the early days of the data sample, the benchmark analyses mostly show that the process is stable. However, the magnitude of nonconforming output is high. On average, approximately 219 of the 806 parts produced each day are nonconforming, corresponding to nearly 272,000 ppm as calculated using Equation 3. The confidence intervals for the mean number of off-spec and bad parts demonstrate that the long-run level of nonconforming output is consistently high, and the prediction intervals show that future daily outcomes are expected to remain similarly excessive without intervention. The Pareto chart (***Figure 11***) shows that a small number of nonconformity types account for the majority of quality problems.

7. Recommendations

These results indicate that improvement efforts should focus not on stabilizing the process, but on reducing the underlying causes of nonconformities. Six Sigma performance (approximately 3.4 ppm nonconforming) is far beyond the current process capability. However, a realistic improvement goal should be established based on client requirements to guide meaningful and achievable quality improvement efforts.

Reducing variation for off-spec-part nonconformance and mistake proofing for bad-part nonconformance in the target categories in order of the Pareto chart (***Figure 11***) first will produce the largest reduction in overall nonconforming output. Button-Hole Diameter, Diffuser-Part Stuck in Mold, and Scroll Wheel Assembly Missing Teeth should be prioritized, followed by the next three most prominent types. This process may be conducted until the process is capable of meeting requirements. Process modifications and improvements should occur between data subgroups with tight statistical process control to effectively monitor the effect of changes and overall process control.